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Experiment 03

Verification of Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL)

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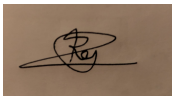
Declaration

I hereby declare that the work presented in this laboratory report entitled **"Verification of Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL)"** is an authentic record of my own work carried out for the course **Basic Electrical and Electronic Circuit (EEE 0713 1206B)**.

This report has been prepared by me independently and has not been submitted elsewhere for the award of any degree, diploma, or credit.

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Objective

- To develop practical understanding of charge conservation (Kirchhoff's Current Law) and energy conservation (Kirchhoff's Voltage Law) in electric circuits.
- To construct multi-node DC networks and accurately record branch currents and potential drops using a digital multimeter.
- To evaluate current distribution at circuit nodes and confirm that loop voltage sums balance to zero.
- To compare empirical measurements against theoretical predictions while evaluating potential sources of physical error.

Introduction

Along with Ohm's Law, Kirchhoff's laws provide the foundational framework required to analyze complex linear resistive networks. Kirchhoff's Current Law (KCL) governs the conservation of electrical charge across circuit junctions, whereas Kirchhoff's Voltage Law (KVL) reflects energy conservation along closed loops. This experiment tests both principles using a multi-branch DC network and quantifies differences between theoretical expectations and laboratory multimeter measurements.

Theory

Kirchhoff's Current Law (KCL)

Kirchhoff's Current Law establishes that the algebraic sum of all currents entering and exiting any node must equal zero:

$$\sum_{k=1}^{n} I_k = 0,$$

taking currents entering a junction as positive and currents leaving as negative. Consequently, total current entering a node matches total current leaving it. For a node where main supply current I divides into two parallel branches:

$$I_{\text{total}} = I_1 + I_2.$$

Kirchhoff's Voltage Law (KVL)

Kirchhoff's Voltage Law establishes that the algebraic sum of all potential drops and rises along any closed loop equals zero:

$$\sum_{k=1}^{n} V_k = 0.$$

When traversing a loop, voltage rises across source terminals are assigned positive values, whereas potential drops across load resistors are negative. For a single loop containing a source V_S and two series resistors, $V_S - V_1 - V_2 = 0$, giving $V_S = V_1 + V_2 = I(R_1 + R_2)$.

Apparatus and Components

Circuit Diagrams

Circuit for KCL Verification

Table 1: List of Equipment and Components Used

Item Description	Specification / Rating	Quantity
DC Power Supply	Regulated DC output	1
Fixed Resistors	R_1, R_2, R_3 (100 Ω–330 Ω range)	3
Digital Multimeter (DMM)	DC Voltage / Current modes	1
Solderless Breadboard	Standard Tie-Point	1
Connecting Wires	Jumper Wires	As required

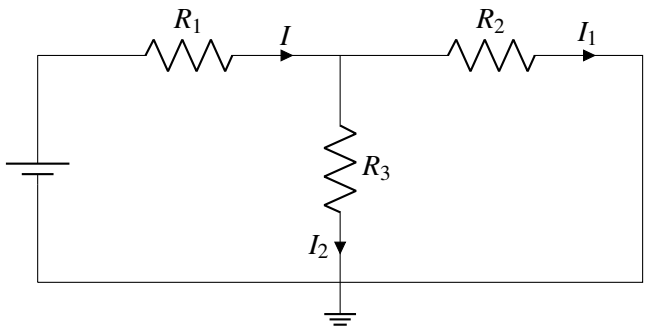


Figure 1: Circuit setup for KCL verification: source current I divides at the main node into parallel branch currents I_1 and I_2, giving $I = I_1 + I_2$.

Circuit for KVL Verification

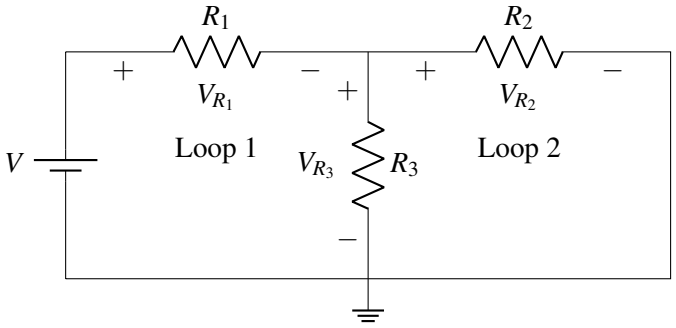


Figure 2: Circuit setup for KVL verification: Loop 1 (R_1, R_3, and V) and Loop 2 (R_2 and R_3) define closed loops.

Experimental Procedure

Part A: Verification of KCL

1. Assemble the network shown in Figure 1 on the breadboard, connecting the DC power supply through R_1 into the junction of R_2 and R_3.
2. Configure the DMM to measure DC current and place it in series with the DC source to record total input current I_{total}.
3. Measure branch currents I_1 and I_2 individually by inserting the DMM sequentially into series with R_2 and R_3.

4. Sum the measured branch currents ($I_1 + I_2$) and compare the total against measured total current I_{total} and nominal theoretical estimates.

Part B: Verification of KVL

1. Adjust the circuit connections to form the dual-loop setup in Figure 2.
2. Set the DMM to DC voltage measurement mode and record potential drops across every resistor (V_{R_1}, V_{R_2}, V_{R_3}).
3. Calculate the algebraic sum of voltages for Loop 1 ($V_{R_1} + V_{R_3} - V$) and Loop 2 ($V_{R_2} - V_{R_3}$) to test agreement with zero.
4. Compare experimental algebraic loops against expected theoretical values.

Data Collection

Table 2: Experimental Data for the Verification of KCL

Input Voltage V (V)	I_1 (mA)	I_2 (mA)	I_{total} measured (mA)	I_{total} theoretical (mA)
10	17.23	9.9	24.4	24.814

Table 3: Experimental Data for the Verification of KVL

Loop	V (V)	V_{R_1} (V)	V_{R_2} (V)	V_{R_3} (V)	Alg. Sum measured (V)	Alg. Sum theoretical (V)
1	10	8.04	1.64	1.64	9.79	10.212
2	10	4.02	0.85	0.85	5.72	5.976

Calculation

KCL Percentage Error

$$\% \text{ Error (KCL)} = \left| \frac{I_{\text{total(theoretical)}} - I_{\text{total(measured)}}}{I_{\text{total(theoretical)}}} \right| \times 100\% = \left| \frac{24.814 - 24.4}{24.814} \right| \times 100\% \approx 1.67\%.$$

KVL Percentage Error

Calculating error across loops yields:

$$\% \text{ Error (KVL)} = \left| \frac{\text{Alg. Sum theoretical} - \text{Alg. Sum measured}}{\text{Alg. Sum theoretical}} \right| \times 100\%.$$

For Loop 1: $|(10.212 - 9.79)/10.212| \times 100\% \approx 4.13\%$. For Loop 2: $|(5.976 - 5.72)/5.976| \times 100\% \approx 4.28\%$.

Result Analysis

The measured total current ($I_{\text{total}} = 24.4\,\text{mA}$) closely matched the calculated value (24.814 mA), yielding an error of 1.67% and verifying KCL. For KVL, loop voltage sums remained within 4.13% and 4.28% of theoretical targets, confirming energy conservation across closed branches.

Discussion and Sources of Error

Minor discrepancies between measured and calculated values stem from specific physical factors:

1. **Ammeter Loading:** Non-zero internal resistance inside the multimeter introduces small burden voltages during current measurements.
2. **Resistor Tolerances:** Discrepancies between nominal color codes and actual resistance values shift measured currents slightly away from theoretical models.
3. **Voltmeter Loading:** Finite internal impedance of the voltmeter draws minor current from measured nodes.
4. **Breadboard Contact Resistance:** Interconnection contacts and jumper leads add small parasitic resistances across loop paths.

Conclusion

Experimental data successfully confirms both Kirchhoff's Current Law and Kirchhoff's Voltage Law within minor error margins. Measured current entering the junction matched the sum of branch currents, and potential changes along closed loops balanced in agreement with theoretical calculations.

References

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